Pokémon Dashboard Project

Purpose

A personal project built for fun and practice — an exploration of how far I could take one idea in Power BI: using real Pokémon game data to build tools that actually help with playing the games better. The goal was figuring out the most efficient way to beat each game — which Pokémon to catch, where, and how they match up against gym leaders — with an eye toward Nuzlocke-style runs in particular, where every catch and every fight matters more because there's no do-over. It's also meant to be published publicly as a demonstration of what I can do with data modeling, Power Query, and DAX.

Dashboards

Dashboard 1 — Pokémon Database

The browsing-and-exploring dashboard. Pick a stat and a matrix lays out every Type against every Generation, so you can see at a glance which types have historically hit hardest. Select any individual Pokémon and a detail panel opens: sprite, classification, flavor text, its evolutionary line (including branching lines like Eevee's), and its full real level-up movepool for whichever game you specify — so you can answer "what does this evolve into" and "what will it actually know at level 30" in one place.

Dashboard 2 — Matchup Optimizer

The problem-solving dashboard, built around the moment you're standing outside a Gym wondering if you're ready. Pick a Gym Leader (or, for Alola, a Trial Captain/Island Kahuna) and it shows their real team for whichever game you're playing. It then ranks every Pokémon you could realistically catch nearby — not just anywhere in the region, specifically near that leader — by how well they'd actually perform, using each candidate's **real, verified moveset** (not a type-based guess) run through genuine dual-type effectiveness math. A breakpoint calculator answers a further question: for a Pokémon you already own, "how much do my odds improve, and at what level, if I keep training it before this fight."

How to Use the Dashboard

Dashboard 1 — Pokémon Database

- Use **Select a Pokémon** to choose one from the list. Its stats appear in **Pokémon Stats**.
- **Learnset by Game & Level** shows every generation/game this Pokémon has appeared in, and the levels at which it naturally learns each move.
- **Evolution Line** shows an image of the selected Pokémon, its pre-evolution, and what it evolves into — for branching lines like Eevee, every possible evolution is shown.
- **Pokédex Entry** shows flavor text about the Pokémon.

Dashboard 2 — Matchup Optimizer

This dashboard helps you beat any gym leader, regardless of which Pokémon you currently have — by telling you what to catch nearby and how strong it'll actually be.

- Use **Select Game & Gym Leader** to pick who you're fighting and which game you're playing.
- **Gym Leader's Team** shows their real Pokémon, levels, types, movesets, and Base Stat Total (BST) — the sum of a Pokémon's six core stats (HP, Attack, Defense, Sp. Attack, Sp. Defense, Speed), used here as a quick measure of overall strength.
- **Best Local Catches** shows every Pokémon you can actually catch near that gym's city or town — routes, wild Pokémon, encounter method, possible levels, the moves they'll have, their BST, and an image. *Real Move Effectiveness* is the strongest damage multiplier this Pokémon's actual, real moveset can land against the selected leader's team — calculated from the specific moves it would really know at that level, not a guess based on type alone. *Candidate Rank Score* is what the table is sorted by: primarily by Real Move Effectiveness, using BST only to break ties, so the best type matchups always rise to the top.
- Click a row in **Best Local Catches** to select a Pokémon. Its sprite appears in **Selected Pokémon**, and **Level-Up Breakpoints** filters to show that Pokémon's complete level-up moveset. *Breakpoint Effectiveness* only shows the levels where training this Pokémon further actually improves its best move against the selected leader — skipping every level-up that doesn't change anything, so you know exactly how much more grinding is worth it.

The Data Wrangling Process

The dashboards are the visible output, but almost all of the actual work happened before a single visual was built — cleaning, matching, and merging more than a dozen source files that were never designed to work together. This section covers that process in some detail, because it's genuinely most of the project.

Building the anchor table: one Pokémon can have four different names

Pokemon_Database.csv became the anchor dimension, but alternate forms (Mega, Gigantamax, regional variants) shared their base name with a separate alt-form field, while two *other* files describing the same Pokémon used two more, completely different naming conventions for the same forms. A CanonicalName column fused these into one unambiguous string, and a dedicated name-bridge table reconciled all three conventions into a shared key for every later merge.

Cleaning the anchor table itself

Text fields were double-quoted at the character level, missing values were the literal string "NULL" rather than a true null, every numeric-looking column had imported as text, and the source file turned out to contain **40 literal duplicate rows across 34 Pokémon** — caught only after a distinct-value count didn't match the row count.

The type effectiveness engine

The source type chart only lists exceptions, leaving every "normal" 1× matchup implicit. It was densified once in Power Query: unpivoted, cross-joined against every type combination, with exceptions merged back in and everything else defaulted to 1×. A second pass caught type combinations recorded in both directions (Grass/Poison on one Pokémon, Poison/Grass on another) — fixed by alphabetizing both types before concatenating.

Parsing multi-row paste-dumps into real tables

Gym leader rosters and route lists arrived as raw copy-pastes of wiki pages, with one logical row spread across 3–7 physical spreadsheet rows. Parsing required detecting the repeating positional pattern rather than treating the file as flat. This surfaced four genuinely different reasons a single gym can have two recorded leaders: **generation** (Koga → Janine), **specific game version** (Wallace vs. Juan), **story progression** (Opal handing off to Bede), and **player starter choice** (Striaton City's Cilan/Chili/Cress trio — eventually split into three real, separate leader entries once it became clear GymLeaderRoster already listed them individually).

Building "what can I catch near this gym" — twice

The first version was assembled from three independent, hand-parsed sources (route adjacency, gym-to-city mappings, and separate Gen 1–6/7/8 encounter spreadsheets) and worked, but was fragile — each source had its own naming quirks. It was later **rebuilt entirely on PokeAPI's own all-generations location and moveset data**, replacing three inconsistent sources with one authoritative one that also added real per-encounter levels and verified movesets for the first time. This migration surfaced a systemic bug worth naming specifically: the water-route naming-correction map wasn't scoped by region, so a real "Water Route 19" in Kanto silently overwrote *every* other region's unrelated "Route 19," incorrectly relabeling ordinary land routes in Kalos and Unova. Fixing it recovered hundreds of rows across three regions at once — a good example of how a single leader appearing to have "no data" (Wulfric, in this case) can trace back to a systemic bug rather than a one-off gap.

Extending the Elite Four

Elite Four members don't belong to a town the way gym leaders do, so "what's catchable nearby" never applied to them — until it was pointed out that every region has some real, walkable location right before its League (Victory Road in most regions, Mount Lanakila in Alola, a plain connecting route in Galar), each with genuine wild encounters that could stand in for "gym city" in the exact same pipeline.

Paldea: a real, if partial, exception to “no data exists”

No bulk Paldea encounter dataset was ever found — but manually sourcing individual Bulbapedia location pages (starting with Glaseado Mountain, covering two gym leaders’ worth of real, level-ranged encounter data) proved the region isn’t a total dead end, just one where the same automation used everywhere else doesn’t scale without real, page-by-page effort.

Matching sprites and movesets without a shared ID

Roughly 258 alternate-form Pokémon had no matching sprite in the original source, requiring direct matching against PokeAPI’s own inconsistent internal conventions — some species’ *default* form needs an explicit suffix (deoxys-normal), a handful default to their *male* variant, and Unown’s 28 letters use an entirely different numbering scheme from every other alternate form in the project. The same category of missing-apostrophe and dropped-accent bugs (farfetchd, flabebe, sirfetchd) turned up independently in the moveset and location data too, and were fixed with a shared name-correction map reused across every source that needed it.

Data Sources & Citations

This project is built entirely on public, community-maintained Pokémon data. Credit belongs to:

- **Bulbapedia** (bulbagarden.net) — gym leader rosters and locations, kahunas, trial captains, route connectivity, and (via direct page lookups) supplementary Paldea encounter data.
- **Serebii.net** — the Generation 1–6 catchable-Pokémon-by-location spreadsheet.
- **PokeAPI** (pokeapi.co / github.com/PokeAPI) — sprite images, species/form reference data, and the all-generations location and moveset datasets that power most of Dashboard 2's real level and moveset accuracy.
- **PokemonDB** (pokemondb.net) — referenced for encounter table structure.
- **Kaggle datasets** for supplementary stat, type, and move reference data (gym leaders, base stats, and move metadata).

All game data is the property of Nintendo/Game Freak/The Pokémon Company. This is a non-commercial, personal data analysis exercise and portfolio piece.

Known Data Quality Findings (disclosed, not silently patched)

- Source-file typos: "Blasoise" for Blastoise, "Camelrupt" for Camerupt, and casing errors ("misty", "roark") in the original gym leader roster.
- The region-unscoped water-route bug described above (Kanto/Kalos/Unova/Alola).
- 40 duplicate rows in the raw anchor source.
- A dropped-apostrophe/accent pattern affecting several species names across multiple independent source files (Farfetch'd, Flabébé, Sirfetch'd, Diglett's Cave, Hau'oli).
- Two location-naming bugs found via direct verification: Glimwood Tangle (Galar) and several Poni Island named-locations (Alola) were never in the base route dataset at all and had to be added from scratch after specific leaders (Opal, Bede, several trial captains) showed zero results.

Known Limitations (explicit, not hidden)

- **Paldea (Gen 9)** has only partial, manually-sourced encounter data (Glaseado Mountain, covering Grusha and Ryme) — the rest of the region has none.
- **Cheren, Marlon, Roxie (Black 2/White 2), and Molayne (Alola)** have no team/roster data anywhere in the sourced files — a genuine gap in the underlying dataset, not something fixable without a new source.
- **A small number of Galar species** (Meowth, Zigzagoon/Linoone, Ponyta) have unresolved base-vs-Galarian ambiguity, since both forms are legitimately wild-catchable at different locations and the current logic doesn't disambiguate by location.
- **MoveReference is missing roughly 65 newer moves** (Gen 8–9 signature moves) not present in its original source.

Build Tooling

- **Python**: one-time heavy data construction (type effectiveness matrix, name-bridge tables, sprite/moveset matching against PokeAPI, encounter-data parsing, the region-aware route adjacency rebuild).
- **Power Query (M)**: ongoing model wiring, merges, and cleaning.
- **DAX**: Field Parameters, role-playing dimensions, matchup/ranking measures, and the cumulative breakpoint-effectiveness calculator.